BrainEx: How Machines Are Redefining Brain Death

The definition for brain death, like many across different medical dictionaries, has evolved with the test of time and advances in technology. Death itself has been divided into circulatory or brain death with key criteria points that need to be demonstrated in order to be determined dead [1]. Challenging established concepts in science has been a driving factor for discovery and innovation, with technological advances pioneering contributors to this scrutiny, definitions must be revisited. BrainEx demonstrates how the criteria for brain death, and the conversations between ethicists and policymakers should be re-examined in order to assess whether they remain appropriate.

Developed at Yale University in 2019, BrainEx is an artificial perfusion system that uses a proprietary haemoglobin-based, acellular, non-coagulative solution called BEx perfusate [2]. This is gently warmed to the normal body temperature of 37ºC and pumped into carotid arteries: mimicking the neural circulatory system. Resulting in 32 pig brains restoring cellular function four hours after their decapitation, these shocking outcomes stimulated the bioethical community [2]. Being able to restore microcirculation, glial inflammatory responses alongside neuronal responsiveness towards events of depolarizing stimulation fundamentally challenges the assumption of irreversibility that is central to the clinical and legal definitions of brain death [2]. BrainEx has prompted bioethical discussion around what we consider brain death, neuroscientific research, as well as the necessary policy that should follow.

The framework for categorising death was first determined by legal standards with the Uniform Determination of Death Act (UDDA) in 1980; it was claimed as either irreversible cessation of cardiopulmonary function or irreversible cessation of all functions of the entire brain, including brain stem [3]. Providing the foundations for diagnosing death through a legislative lens, the UDDA faced criticism by philosophers and doctors before and beyond the passing of this definition [4]. This meant that in 1995, the American Academy of Neurology (AAN) published the first formal clinical criteria that would define the standards of determining brain death from a medical perspective. It considered "[t]he three cardinal findings in brain death [to be] coma or unresponsiveness, absence of brainstem reflexes, and apnea. [5]" However, in attempting to standardize death, both guidelines ultimately uncovered how it was only possible to approximate how death was determined.

In 2023, a multi-society consensus that included the AAN established a guideline for what is currently accepted as the medical standard of death, on the basis of whole brain formulation [6]. Yet, the distinction between brain activity and brain function persisted in ambiguity; with the terms being used interchangeably in scientific literature [7]. This lack of clarity has led technology like BrainEx to take advantage of a lingering grey zone, with their distinction perhaps being critical terms in defining death and its regulations. While this technology contributes to the bioethical conversations around brain death, neuroscientific scrutiny regarding these topics has been contested long before.

What binds brain death most confidently is the permanence in cessation of brain function, this irreversibility being imperative to both legal and clinical distinctions of death. Having BrainEx demonstrate impermanence by restoring cellular and molecular function, hours after what would be considered irreversible death, destabilises a foundational dogma and the assumptions of the UDDA [3]. This could propose the idea of brain resuscitation; having implications for brain-dead patients who are eligible for organ donations. If the possibility of restoration persists, physicians and families of these patients could urge further intervention. The results that BrainEx has produced could exponentially increase the gap between potential donors and transplant recipients with the hope of restoration being extrapolated. While scientists at Yale were careful in claiming that none of the brains regained any semblance of organized electrical activity that brings about consciousness, ethicists rapidly raised questions on how this may affect organ donation from people declared brain dead, as well as the possibility of undermining the dead donor rule [8].

A component of the solution circulating in the 32 pig brains was lamotrigine: an anti-seizure drug that works by either dampening or completely blocking neuronal activity, used under the impression that brain cells would be better preserved [9]. However, when the solution was washed off of a few single cells, electrochemical responses appeared. The lack of dismissal towards global electrical activity instigating consciousness can only suggest that conversations around consciousness are not out-of-reach

[9]. The precautionary principle of also administering anaesthesia also proposes the possibility that signals could be consciousness-associated. BrainEx has also created the possibility of a new scientific limbo; balancing their research on the border between deceased and living animal tissue as partially reactivated post-mortem brain investigation. Ultimately, the pace of progress from BrainEx has stimulated an incredible curiosity around what was previously thought of as an irreversible determination of death.

In order for bioethical, clinical, legal and policymaking actors to address the possibility of such challenges, there must be a reconsideration for “irreversibility” in the UDDA. Scholars have brought up the necessary revision of the UDDA’s for both clinical and practical distinction between contextual irreversibility and biological irreversibility [10]. Currently, there is an active reexamination of the most defensible brain criterion between the whole brain, brain-as-a-whole, higher brain, or brain storm in order to appropriately define death by neurological criteria [11]. This reorganization considers the way death is defined beyond biological confines, as a social, religious, and philosophical concept. Ultimately, it has been proposed to be explicit in defining the permanence from that founds the UDDA’s definitions as "without currently available or reasonably foreseeable treatment," in order to accommodate for a fast paced evolution of scientific and medical possibility [12].

Furthermore, the research that led to BrainEx has uncovered the need for a framework of postmortem brain experimentation. Intervention for simple cellular restoration must be separated from restoring integrated organ function without consciousness, as well as when postmortem brain intervention approaches semblance of consciousness. Considering a tiered division between intentions of research can ensure that the appropriate informed consent protocols, ethical oversight and regulatory frameworks are in place while promoting scientific innovation at a digestible pace. In addition, considering the sensitivity around organ donation and the possibility that BrainEx research has suggested, the need for organ donors will further increase. Hence, I suggest a universal opt-out protocol for organ donation as seen by Spain, Austria, France and further countries across Europe and Latin America. Having presumed consent for organ donation will confidently address the gap between donor and recipient, as seen with a systematic review of Opt-out Versus Opt-in consent on deceased organ donation and transplantation between 2006 and 2016 [13]. Updating organ donation protocols whilst introducing the possibility for a formal and distinct brain resuscitation candidacy assessment can help preserve trust in the organ donation system while also addressing the persistent demand for organ donation.

Ultimately, there has been a constant race between scientific innovation and ethical conversation and BrainEx is no exception, nor trend setter. It simply slots itself in the gap between progress and moral frameworks. Conversations on bioethical subject matter and the technology that challenges them must be consistent and invasive in order to remain relevant. International and interdisciplinary discussion will engage with cutting edge science in a way that promotes its curiosity under the guidance of ethical oversight. This has also fueled the argument separating brain activity and life, with the brain being an integrator of the body, and cellular activity amiss of integration does not result in life claimed by Catholic philosophers [14].

BrainEx has revealed that the boundary between what we consider to be life and death is still far away from being clear or distinct. Claiming for a threshold has proved to be contingent on the current technological breakthrough, shifting with each novelty. However, the high stakes regarding what is determined to be brain death has led for decisions around end-of-life, organ donation, and research ethics to face an increasing level of mistrust, eroding credibility of the medical professionals. The discovery that has flourished from Yale has stimulated a necessary call to action for legislatures, ethical actors, and clinical safeguards to evolve alongside emerging neurotechnologies. In order to bioethically keep up, discourse must be provoked. BrainEx is not the first to challenge central dogmas, and it won’t be the last.

References:

[1] Machado, C. (2010). Diagnosis of brain death. Neurology international, 2(1), e2.

[2] Vrselja, Z., Daniele, S. G., Silbereis, J., Talpo, F., Morozov, Y. M., Sousa, A. M., ... & Sestan, N. (2019). Restoration of brain circulation and cellular functions hours post-mortem. Nature, 568(7752), 336-343.

[3] Bernat, J. L. (2023). The brain-as-a-whole criterion and the Uniform Determination of Death Act. AJOB neuroscience, 14(3), 271-274.

[4] Bernat, J. L. (2023). Challenges to brain death in revising the Uniform determination of death Act: the UDDA revision series. Neurology, 101(1), 30-37.

[5] Quality Standards Subcommittee of the American Academy of Neurology. (1995). Practice parameters for determining brain death in adults (summary statement). Neurology, 45, 1012-1014.

[6] Greer, D. M., Kirschen, M. P., Lewis, A., Gronseth, G. S., Rae-Grant, A., Ashwal, S., ... & Halperin, J. J. (2023). Pediatric and adult brain death/death by neurologic criteria consensus guideline: report of the AAN Guidelines Subcommittee, AAP, CNS, and SCCM. Neurology, 101(24), 1112-1132.

[7] Raichle, M. E. (2010). Two views of brain function. Trends in cognitive sciences, 14(4), 180-190.

[8] Truog, R. D., & Robinson, W. M. (2003). Role of brain death and the dead-donor rule in the ethics of organ transplantation. Critical care medicine, 31(9), 2391-2396.

[9] Greenfieldboyce, N. (2019). 'Mind-blowing' ethical issues arise as brains of dead pigs are partially revived. KQED.
https://www.kqed.org/science/1940480/mind-blowing-ethical-issues-as-brains-of-dead-pigs-are-partially-revived

[10] Bernat, J. L. (2023). Challenges to brain death in revising the Uniform determination of death Act: the UDDA revision series. Neurology, 101(1), 30-37.

[11] Robbins, N. M. (2023). What is the ideal brain criterion of death? Clinical and practical considerations: the UDDA Revision Series. Neurology, 101(2), 83-85.

[12] Pérez, A. M. (2022). Brain death debates: From bioethics to philosophy of science. F1000Research, 11, 195.

[13] Ahmad, M. U., Hanna, A., Mohamed, A. Z., Schlindwein, A., Pley, C., Bahner, I., ... & Jarmi, T. (2019). A systematic review of opt-out versus opt-in consent on deceased organ donation and transplantation (2006–2016). World journal of surgery, 43(12), 3161-3171.

[14] Berendes, S. (2024). Moral Certitude in Brain Death Diagnoses in Light of OrganEx and BrainEx. National Catholic Bioethics Quarterly, 24(1).